Peijie Liu

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Objective

Passionate Electrical Engineering sophomore with extensive PCB layout experience and full-stack electronics development skills. Avionics Reliability Lead & HITL Responsible Engineer at YJSP. Particularly adept with operational amplifier applications, precision and high-speed ADC/DAC systems, and FPGA HDL programming. Seeking an internship in hardware design starting May 2026.

Education

Georgia Institute of Technology | Atlanta, GA *August 2025 – Present*
Bachelor of Science in Electrical Engineering, Minor in Robotics, GPA 4.00. *Expected Graduation, Dec 2028*

Skills

Hardware: STM32/AVR MCUs, Artix-7/ZYNQ FPGAs, Analog Components, Oscilloscope, Logic Analyzer.
Software: Altium Designer, LTSpice, PSpice, Vivado, ModelSim, STM32CubeIDE, IntelliJ, VS Code, MATLAB.
Programming: C++, C, RISC-V Assembly, Verilog HDL, MATLAB, Java. Adept with Free RTOS, HAL, & DMA development.
Languages: English (fluent), Mandarin (native), Japanese (beginner); Communication via conference, end-users, and live streaming.

Experience

Yellow Jacket Space Program | Georgia Institute of Technology *August 2025 – Present*
Avionics Reliability Lead & Hardware-In-The-Loop (HITL) Responsible Engineer
- Design, build and operate the HITL system to validate full avionics integration against a real-time emulated rocket. Led hardware & software teams to develop automated CI/CD pipeline. Delivered zero critical failure across our record-setting KeroLox system.
- Architected precision DAC system with analog backends and ADC self-calibration to accurately emulate RTD, Valve, PT, TC, etc.

RoboRambler | Georgia Institute of Technology *January 2026 – Present*
Electrical Control Lead
- Revitalized CAN–UART–USB protocol converter design; developing supercapacitor multiphase Buck-Boost intelligent converter.

Interactive Music Group – Vertically Integrated Projects | Georgia Institute of Technology *January 2026 – May 2026*
DSP Engineer
- Developed AI-powered self-meshing DSP system and adaptive centroid MPE matrix for interactive musical instruments.

Engineering Intern | Moondrop Co., Ltd. *September 2024 – August 2025*
Technical Content Creator & Contracted Vlogger
- Curated channel that attracted 1M+ views and 10k+ active subscribers, generating consistent monthly revenue of $200.
- Conducted electroacoustic research on DAC circuits, achieving industry-leading performance of -145dB THD and -123dB THD+N.

Projects

µCHIMERA(Micro Composite Harvesting & Integrated Modular Energy Regeneration Array) - *Chief Engineer* *January 2026*
*A stackable multi-source energy harvesting system for battery-free or implantable wearable, integrating **Solar** (5µW-510mW with MPPT), **Thermoelectric** (cold-start at 20mV, operates at 0.2°C ΔT), and **Piezoelectric** (88mW RMS from random vibration) modules.*
- Engineered power conversion circuitry featuring <1µA quiescent current and 4-ch stackable energy routing with smart control.
- Competed in Georgia Tech's 2026 InVenture Prize, securing $1,000 in seed funding for advancing energy-autonomous technique

Super FDA (Fully Differential Amplifier) - *Individual Product Manager and Developer* *April 2025*
A 200W fully differential Class AB amplifier that can shift dynamically among 4 feedback architectures and 2 input stages, achieving as much -135dB THD and -116dB THD+N, realizing 80Vpp output under ±24V supply, and supporting stable remote feedback.
- Engineered innovative Mosfet bias compensation circuit that reduced crossover distortion, achieving -105dB open-loop THD+N.
- Developed high-power linear LDO architecture that achieves 20A output with 2.5V dropout voltage. Patent application pending.
- Designed and fabricated 5 PCB iterations, successfully launched product to market and generated over $1,000 in revenue.

Relevant Coursework

Circuit Analysis: DC/AC circuit theory, operational amplifiers circuits, network analysis techniques (node/mesh analysis, Thevenin/Norton theorems), phasor analysis, frequency response, Bode plots, Laplace transforms, power/transient analysis.

Programming for Hardware/Software Systems: C programming, RISC-V & MIPS assembly, data structures, memory management, operating systems, embedded systems, performance optimization, compiler design, ARM microcontroller programming.

Awards

IEEE Robotech 2026 | Georgia Institute of Technology | "Tachy-Astroach" *January 2026*
Champion Team | Chief Electronic Engineer
Designed and built full electronics hardware for our remote-controlled parent-subunit quadruped robot system within 36 hours.